

What Should World Models Forget?

Stratified Retention for Continual Adaptation

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Abstract

Continual learning treats degradation on previously seen data as evidence of failure, a convention inherited from settings with a stationary prediction target, where a correct label remains correct indefinitely. World models do not satisfy this condition. Their prediction target is the environment, which changes, so knowledge that was accurate when acquired may later become false, and discarding it is required behavior rather than a defect. Non-stationary ground truth is well studied in the concept drift literature and in the temporal factuality of language models, but has not been formulated for world models, which are distinctive in that they also encode knowledge that must never be revised. We argue that continual world models require retention stratified by *invariance timescale*, separating invariants such as physics and object permanence, which must never be revised, from instance-level facts that should be revised as soon as the environment changes. Standard forgetting metrics cannot distinguish a world model that has correctly revised outdated knowledge from one that has suffered catastrophic forgetting, and consequently rank a frozen model highest, while existing physical-reasoning benchmarks evaluate only frozen checkpoints. We propose *differential retention*, which reports invariant regression testing across the adaptation stream jointly with revision latency, without aggregation.

1 Introduction

World models learn an executable approximation of environment dynamics, mapping a state and an action to a distribution over successor states, and are used to plan, to train policies in imagination, and to synthesize experience that would otherwise be costly or unsafe to collect. Recent systems have scaled this formulation considerably. Dreamer 4 trains agents entirely within a learned model of Minecraft from offline data alone [10], while V-JEPA 2 pretrains on over one million hours of video and transfers to zero-shot robot manipulation by planning in a learned latent space [2]. Pretraining efficiency has improved in parallel, with LeVJEPa matching V-JEPA 2 at between $5.6\times$ and $20.8\times$ less total pretraining compute [14]. In each case the resulting model is trained once and deployed without further modification.

The environments these models operate in do not stay fixed. A deployed system must detect when its predictions fail, absorb new evidence, and revise its account of the world. Continual learning supplies the tools for that revision, but rests on an assumption that world models do not satisfy: that the prediction target is stationary, so any decline in performance on previously learned material is a defect. That convention derives from supervised classification. An image of a cat continues to be an image of a cat, so a model that becomes less accurate on it has degraded. A world model instead approximates the dynamics and configuration of an environment, $p(s_{t+1} \mid s_t, a_t)$, the distribution over the next state s_{t+1} given the current state s_t and the action a_t taken at time t . That environment

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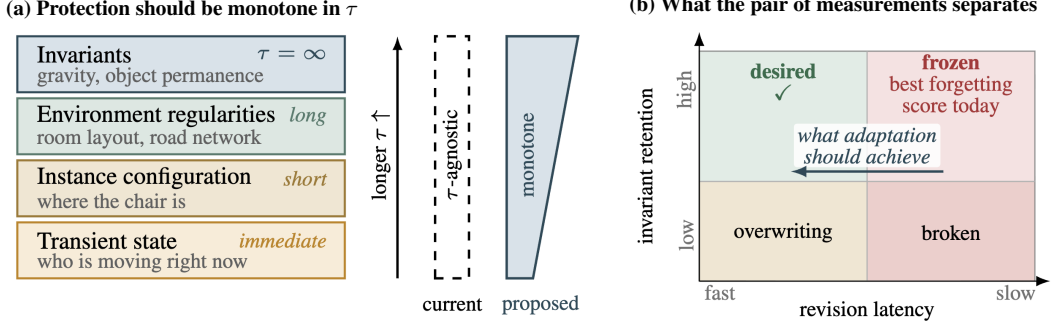


Figure 1: (a) World-model knowledge stratified by invariance timescale τ . Existing consolidation methods choose what to protect by importance to past performance, which is unrelated to τ , whereas we argue protection (bar width) should increase with τ . (b) Reporting invariant retention and revision latency jointly helps separate a model that has adapted correctly from one that is merely frozen. Standard forgetting metrics reward both high retention and slow revision, and are therefore maximized by the frozen model in the top-right quadrant.

is not fixed. When a road is reconstructed, an object is relocated, or a scene is rearranged, knowledge that was accurate at acquisition becomes false, and a model that continues to assert it is stale rather than faithful. This separates three phenomena that current practice reports as one. **Catastrophic forgetting** is the loss of knowledge that remains true, and is a defect. **Deliberate unlearning** is the removal of knowledge that remains true but is unwanted, and is well studied. **Obsolescence** is the retention of knowledge that has become false, where revision is mandatory.

The concept drift literature is built on the premise that ground truth can change. Ashrafee et al. [1] observe that continual learning methods implicitly treat the data of earlier tasks as static. In language-model factuality, Jiang et al. [11] propose the Evaluation Misleading Rate, which counts how often a model gives an up-to-date answer and is marked wrong against an outdated benchmark label. The problem has not, however, been posed for world models, and does not carry over unchanged. In both settings above, nothing the model knows is permanently true, so any fact may be revised. A world model is different: the physical, geometric and causal structure it encodes should survive every update. World models are therefore the case where obsolescence coexists with hard invariants, and that combination changes what must be protected and what must be measured (Figure 1).

Contributions. We argue that obsolescence is a failure mode of continual world models distinct from catastrophic forgetting, and one that invariants make unavoidable (Sec. 2). We propose stratifying world-model knowledge by invariance timescale τ , with protection increasing in τ (Sec. 3). We then identify what current evaluation cannot measure (Sec. 4) and propose differential retention as a protocol that addresses it (Sec. 5).

2 World models break the stationary-target assumption

A world model can be wrong about the past for two reasons, and its behavior alone will not tell you which. It never acquired the relevant knowledge, or it acquired that knowledge and the environment subsequently changed. Only the first constitutes a learning failure, yet the two are reported identically.

The continual world-model literature has largely not been required to confront this, because its benchmarks are constructed such that the second case cannot arise. Kessler et al. [12] evaluate Continual-Dreamer on sequences of Minigrid and Minihack tasks, and Yang et al. [23] evaluate WMAR (World Models with Augmented Replay) on Procgen and Atari sequences. In both cases ground truth within a task is fixed throughout, so any subsequent degradation is genuinely forgetting. Sometimes the assumption is explicit. DRAGO [9] studies task sequences that share the same dynamics and differ only in reward. This is a reasonable simplification, but it is also exactly the setting in which knowledge can never go out of date.

Where environments do change, world models accommodate the change poorly. Wan et al. [21] report that PlaNet and DreamerV2 adapt poorly to local changes in the environment. Nijjer [18]

Table 1: Strata of world-model knowledge ordered by invariance timescale τ .

Stratum	τ	Examples	Required behavior
Invariants	∞	gravity, object permanence, impenetrability, geometric consistency, action–effect structure	never revised, as any change is a defect
Environment regularities	long	room layout, road network, how other agents typically behave	revised slowly, on accumulated evidence
Instance configuration	short	where objects are, which door is open, current lighting	revised promptly
Transient state	immediate	who is moving where, momentary occlusion	no retention expected

observes that in DreamerV3 with an unbounded replay buffer the world model retains essentially all measurable knowledge of earlier tasks while the policy degrades regardless, suggesting that where a world model fails to update, the mechanism need not be loss of stored information. The gap is therefore structural rather than accidental: the setting where obsolescence matters most is the one our benchmarks leave out.

3 Stratification by invariance timescale

We propose organizing world-model knowledge not by task or by modality but by τ , the timescale over which a given piece of knowledge remains true. τ is a property of the environment, not of the model or its update schedule.

Table 1 expresses a claim that is not reflected in current practice. Elastic weight consolidation [13], replay [19], and distillation-based consolidation [15] do decide what to protect, but they decide on the wrong basis: what was important for past performance, rather than how long that knowledge stays true. The two are unrelated. A parameter can be important for an earlier task while encoding a fact the environment has since invalidated, and these methods respond by protecting it most strongly.

The result fails in both directions. Nothing in a standard continual pipeline checks that physical or geometric invariants survived an update, so the top stratum is under-protected. Meanwhile a replay buffer that faithfully stores the old layout of a room keeps reasserting it after the room has changed, so the lower strata are over-protected. Ashrafee et al. [1] fix the second problem in classification by evicting outdated exemplars. The world-model case also needs the first, which has no counterpart in their setting.

LeJEPa [3] identifies the isotropic Gaussian as the embedding distribution that minimizes downstream prediction risk and enforces it with a regularizer that provably rules out representational collapse, an objective LeVJEPa extends to video [14]. No comparable proof exists that a representation *will* preserve a physical or geometric invariant once it starts updating. The available formal results describe what a representation must *not* do, but none describes what it must retain.

This stratification is also distinct from multi-timescale memory. The Continuum Memory System of Behrouz et al. [4], building on Titans [5], stratifies memory modules by update frequency, whereas ours stratifies by the rate at which ground truth moves. Only the latter tells you whether a change was correct: a fast-updating module that revises a physical invariant is behaving exactly as designed, and is still wrong, because the invariant it revised did not change. The top stratum is not only physics. Object permanence, geometric consistency, causal action–effect structure, and the agent’s model of its own body are all knowledge that should not be altered. Physics is simply the part for which probes already exist.

4 What current evaluation cannot measure

Standard continual-learning metrics such as backward transfer and the forgetting measure compute performance on an earlier task before and after subsequent learning and interpret any decrease as forgetting. They cannot see the distinction drawn in Sec. 2. A model that degrades because it has broken and a model that degrades because it correctly revised knowledge the environment has since invalidated produce the same number. Díaz-Rodríguez et al. [7] argued that forgetting alone

is insufficient for evaluating continual learners and proposed axes covering transfer, memory and efficiency. None of them asks whether a degradation was warranted.

This creates a perverse optimum: the best possible forgetting score belongs to a model that never updates at all. In a stationary benchmark this is harmless, since such a model also performs poorly on subsequent tasks. In a changing environment it is not, because the frozen model is precisely the failure mode continual adaptation is intended to remedy. A second gap follows. In classification and textual factuality the metric can be repaired by refreshing the labels, but in world models this is insufficient, because matching a refreshed target says nothing about whether the invariants survived the update. A model that accommodates a rearranged scene by degrading its representation of object permanence would score well under any updated-target metric.

5 Differential Retention

We propose evaluating a continual world model on two quantities, reported separately.

(a) Invariant regression testing. We fix a probe set drawn from the $\tau = \infty$ stratum, held out from the adaptation data, and re-run it after every adaptation round. A single evaluation at the end cannot show which update damaged an invariant. Concept-isolated physical probes already exist and are directly reusable. WorldBench [20] disentangles individual physical concepts to support per-concept diagnosis, and PhyGround [16] covers thirteen physical laws spanning solid-body mechanics, fluid dynamics and optics. Both, as published, evaluate frozen models. Applying them as a regression suite across an adaptation stream is, to our knowledge, not done in the literature, and requires no new data, only a change in when the probes are run.

(b) Revision latency. We measure how many observations the model needs before its predictions reflect a fact that has changed, and count separately the facts it never revises at all.

Neither quantity is informative on its own, as Figure 1(b) shows. A frozen model scores well on invariant retention and fails on revision latency, while a model that adapts by overwriting scores well on revision latency and fails on invariant retention. Only a model that satisfies both is doing what continual adaptation is meant to achieve, and no single number can capture that.

Maes et al. [17] report that recoloring the background in Push-T reduces planning success from 50.8% to 6.0% for both LeWM and PLDM, and conclude that current world models remain brittle under distribution shift. Were such a model permitted to adapt online and observed to recover, current evaluation cannot determine whether recovery occurred through correct revision of an appearance prior or through degradation of the dynamics model into a form that happens to score well on the task at hand. We note that (a) is implementable today only for the physical sub-case, since comparable probes for geometric consistency and causal structure do not yet exist.

6 Related Work

Continual world models. Kessler et al. [12] find that reservoir sampling over past experience robustly mitigates forgetting, and adopt it as the replay strategy of Continual-Dreamer. Yang et al. [23] make the buffer memory-efficient by matching the distribution of past experience, and Fu et al. [9] avoid storing data at all by rehearsing synthetic trajectories. All three assume a fixed target. **Non-stationary ground truth.** The concept drift literature supplies the framing we adopt [1, 6], and Jiang et al. [11] build the closest existing metric, though for language-model factuality. **Evaluating world models.** Upadhyay et al. [20] and Lin et al. [16] isolate individual physical concepts so that failures can be attributed to specific laws, Maes et al. [17] document how sharply performance falls outside the training distribution, and Xing et al. [22] argue more broadly about what a world model is for. **Memory and plasticity.** Behrouz et al. [4, 5] stratify memory by update frequency, which we contrast with ours in Sec. 3. Dohare et al. [8] show that networks lose the capacity to learn over long streams, a constraint on any continual scheme.

7 Limitations and open questions

As a position paper, this work contributes a framing and an evaluation protocol rather than experimental results, and we identify three questions that the framing raises. First, the strata in Table 1 are a

starting proposal, their boundaries are likely domain-dependent, and whether τ can be estimated from observation or must be supplied as domain knowledge remains unclear. Second, probes of sufficient rigor exist today only for physical invariants, so realizing the protocol in full requires comparable probes for geometric consistency and causal structure. Finally, whether invariants are localized in a learned world model is not yet known, and if they are entangled with its other knowledge, stratified retention may need architectural support.

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